Two geographies of urban deprivation: everyday services and emergency care across European cities

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Abstract. Proximity has become the planning paradigm of European cities: the 15-minute city promises every daily need within a short walk and is advanced as a route to urban resilience. That promise concerns everyday life. Urban populations also face a growing burden of emergencies, with heat alone killing an estimated 44,000 Europeans a year and the number exposed to extreme heat projected to rise from ten million to almost 300 million a year under three degrees of warming; in an emergency what matters is whether care reaches people in time. Accessibility research measures both kinds of need as travel time, which cannot say how much a resident loses when a service lies further away. Deprivation theory, developed in humanitarian logistics to quantify the welfare lost during time without an essential good, can. We bring that theory into urban analysis and ask whether the deprivation cities impose follows the same law for everyday and for emergency needs. Measuring both on a 100 m population grid for 67 functional urban areas in 24 European countries, we find two regimes that obey different laws. Everyday deprivation falls with city size while emergency deprivation does not, and everyday inequality widens with size while emergency inequality never narrows. Emergency deprivation follows the coverage of national emergency systems rather than city size: five cities that appear average by travel time carry 2.4 to 4.7 times the typical emergency deprivation. Within cities, the neighbourhoods most deprived of everyday services are also, more often than chance, the most deprived of emergency care, and neighbourhoods with many children carry this compounded deprivation most consistently. A city that works every day is not thereby resilient.

Significance. Cities are planned for everyday proximity, yet urban populations increasingly face emergencies in which minutes decide survival. We show that everyday and emergency needs form two regimes of urban deprivation that obey different laws. Measuring anchored deprivation rather than travel time across 67 European city regions, larger cities reduce everyday deprivation but not emergency deprivation, everyday inequality grows with city size, and emergency deprivation follows national coverage, with five capitals harbouring emergency deserts that travel-time averages cannot detect. The same neighbourhoods, often those with many children, lose in both regimes. A city that works every day is not thereby resilient; urban resilience requires both regimes.

Keywords: urban deprivation; accessibility; 15-minute city; emergency medical services; urban resilience; spatial equity; urban scaling

Introduction

Three quarters of Europeans live in urban areas, a share projected to exceed 83 % by 2050 [1], and the cities they live in are increasingly planned around one idea: that every daily need should lie within a short walk. The 15-minute city has become the planning paradigm of the continent [2, 3], and it describes an everyday regime, in which residents choose when to reach a general practitioner, a pharmacy, a supermarket, a school or a park. Urban life is also lived in a second regime. Heat kills an estimated 44,000 Europeans a year, a toll projected to triple by 2100 under current policies [4], and the number exposed to extreme heat rises from about ten million a year today to almost 300 million under three degrees of warming [5]. The emergency regime is exercised every day and not only in disasters: across Europe about 8 % of people who suffer a cardiac arrest outside hospital survive, the ambulance reaches them after 12 minutes on average, and each minute of delay lowers the probability of survival [6–8]. The European Union now asks member states to plan for emergencies explicitly [9]. Everyday life and emergencies place different demands on the same city. What does a resident lose when the services they need lie further away, and does the answer differ between the everyday and the emergency regime?

Urban research has one established answer, and it is a measurement rather than a theory. Accessibility, defined by Hansen as "a measurement of the spatial distribution of activities about a point" [10], is operationalised as travel time. The 15-minute-city programme [2, 11], global travel-time mapping [12, 13] and the measurement of unequal access within cities [14] all count minutes, and a minute counts the same wherever it falls, for whichever service, for whichever resident. Combined with urban scaling, whose empirical regularity [15] and generative theory [16] hold that infrastructure and services improve systematically with city size, the access position implies

a prediction: larger cities should serve every need better, everyday and emergency alike. The position has no reason to expect two service classes to behave differently.

Welfare economics offers a second position. Deprivation theory, developed in humanitarian logistics, defines the deprivation cost of a good as the welfare lost during the time a person spends without it, and shows that this cost is estimable as a function of deprivation time [17, 18]. Two objects follow, and their difference carries this paper. For substitutable goods the loss is a *level*: bounded and S-shaped, saturating once people adapt and substitute [19]. For critical goods the loss is a *cost*: convex and unbounded, rising ever more steeply with time, with estimates for goods and for emergency ambulance services [20–22]. The value of time of transport economics, estimated on routine trips whose delay can be rescheduled, is the wrong valuation for the second object: the clinical evidence that survival after cardiac arrest falls by one to two percentage points per minute of delay [8, 23] is a statement about mortality. Deprivation theory therefore predicts what the access position cannot: because the welfare lost follows the need, the two regimes need not obey the same law, and a city may serve one while failing the other. Urban resilience research reaches the same distinction from the other side: resilience is defined through equitable access to essential services [24, 25], the 15-minute city is advanced as a resilience strategy [2], yet everyday vulnerability profiles fail as predictors of crisis-time capability [26], and whether the neighbourhoods a city fails every day are also those it fails in an emergency [27] is an open empirical question.

The stakes are not academic. By the standard access statistic, the median drive time to emergency care, Bucharest is served better than the typical European city (8.5 minutes, 0.71 times the sample median). By deprivation cost it carries 2.8 times the typical burden, because a well-served core pulls the median down while a deep periphery lies far beyond any clinical response benchmark. Which number a planner sees decides whether Bucharest has an emergency problem at all. In this paper we measure which of the two positions describes urban Europe.

We do so for 67 functional urban areas (FUAs) in 24 European countries, from 101,050 to 12.9 million inhabitants, drawn by a stratified design over four macroregions and four size strata, on a common 100 m population grid. Europe is the setting for three reasons. It has a harmonised functional city definition [28], a harmonised census grid, near-complete street and facility mapping, and clinical emergency-response benchmarks shared across national systems [29–31], which global travel-time mapping [12, 13] cannot carry. It is the mature end of the urban life cycle in which large cities lose their growth advantage [32], so its cross-sectional size gradient reads what size delivers once growth no longer favours the large city. And access to essential services is a right in the European legal order [33, 34], with emergency care organised nationally under shared response targets, so the sample covers one polity with one set of obligations and 24 national emergency systems. Following the scaling tradition we estimate log-log elasticities of city outcomes on population [15]; every size gradient is cross-sectional, and "as cities grow" means the space-for-time reading of that gradient, never a longitudinal observation.

Three findings follow. First, deprivation is a different object from access: it explains the cross-city structure better than travel time, agrees with travel time wherever results are rank-based, and reveals what travel-time averages hide, five emergency deserts in European capitals. Second, urban growth delivers on one clock only: everyday deprivation falls with city size and emergency deprivation does not, everyday inequality widens with size and emergency inequality never narrows, and emergency deprivation follows the coverage of national emergency systems rather than city size. Third, the two deprivations compound: in 66 of 67 cities the neighbourhoods most deprived of everyday services are also, more often than chance, the most deprived of emergency care, and neighbourhoods with many children carry the compounded burden most consistently.

Results

Two surfaces summarise each city (Fig. 1; Materials and Methods). The *everyday deprivation level* of a grid cell, between 0 and 1, expresses how far the burden of reaching five everyday service categories on foot has progressed toward saturation: near 0, all services are comfortably walkable; near 1, additional minutes no longer add burden. The *emergency deprivation cost* of a cell is an open-ended positive number expressing the welfare lost through its drive time to the nearest emergency facility, in multiples of the loss at the 15-minute emergency-medical-service (EMS) benchmark: a cell at 2.0 carries twice the deprivation of one whose ambulance arrives at the benchmark, and the cost keeps escalating beyond it. "Cost" is welfare lost, never money, and deprivation throughout is the welfare loss imposed by poor physical access, distinct from the socioeconomic deprivation of composite indices, which enters only as an explanatory variable. The two objects differ because the needs do: everyday services can be substituted or deferred, so the marginal minute loses value once coping sets in and the loss is bounded; emergency care cannot, survival worsens with every minute, so the marginal minute gains value and the loss is unbounded. The linear loss of a travel-time average over-prices minutes where they are welfare-inert and under-prices them where they are clinically decisive. One discipline follows: only unit-invariant statistics (elasticities, Ginis, ranks) cross regimes, and raw magnitudes never do. Table 1 summarises the sample. Cities nest in

Table 1: Sample by macro-region: 67 functional urban areas (Eurostat/OECD definition), stratified over 4 regions × 4 size strata, 113.9 million residents. $\bar{D}_{\mathrm{ev}}$: mean everyday deprivation level; $\bar{C}_{\mathrm{em}}$: mean emergency deprivation cost (multiples of g(15); not comparable in magnitude); ρ: rank coupling of the two surfaces; HH: compounding share at the median split. Full per-city table in SI Appendix.

Region	Cities	Countries	Pop. (M)	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	ρ	HH
North	13	4	11.3	0.41	1.81	0.47	0.34
West	16	5	37.1	0.23	0.74	0.36	0.32
South	18	4	39.8	0.29	1.07	0.38	0.32
CEE	20	11	25.6	0.30	1.48	0.55	0.36
All	67	24	113.9	0.29	1.14	0.42	0.33

24 countries, so every scaling claim cites a wild cluster bootstrap p-value on country clusters and every regional contrast a country-permutation p-value.

Deprivation is a different object from access

Every headline regression is estimated twice, on the deprivation surfaces and on the minutes-based indicators the accessibility literature would use [13, 14], and the comparison has three parts (Fig. 2; exact values in SI Appendix). Deprivation is the better-behaved object: city size explains 44 % of the cross-city variance in mean everyday deprivation against 16 % for the median walk minute, because the saturating level function discounts the variation that carries no welfare content (whether a periphery cell sits 50 or 70 minutes from a pharmacy moves a travel-time average and no household's routine). The filtering costs nothing in ordering: the deprivation-based and time-based Gini families rank cities almost identically (Pearson $r = 0.965$ everyday, 0.976 emergency), which is why every rank-based result below is parameterisation-free and why the level and inequality results are where the calibration carries information.

Deprivation reveals what averages hide. Bucharest's median emergency drive time is 8.5 minutes, 0.71 times the sample median, so by the standard access statistic the city is better served than typical urban Europe; its mean emergency deprivation cost is 2.81 times the sample value (Fig. 2B). Vilnius, Ljubljana, Rīga and Tallinn show the same anatomy, median-time ratios of 0.71–1.92 against deprivation ratios of 2.45–4.72: a well-served core pulls the median minute down while a deep periphery, far beyond the 15-minute benchmark, carries convex costs no average over uniform minutes can register. These five are the emergency deserts below. Even inside the access paradigm the signal sits in the tail: the only minutes-based emergency statistic to reach significance in the size regressions is the 90th-percentile drive time (-0.059, $p = 0.017$).

Deprivation agrees with access wherever the result is about ranks, which protects the findings below from being calibration artefacts. The everyday size gradient is present in plain minutes (median walk-time elasticity -0.237, $p = 3.8 \times 10^{-5}$) and the absence of an emergency gradient is present in minutes too (-0.043 to -0.059 across median, mean and 90th-percentile drive time). No headline depends on the deprivation functions; they restate in welfare terms a structure the network geometry carries, and add the levels, the inequality magnitudes and the tails.

Larger cities have lower everyday deprivation, but not lower emergency deprivation

If the returns to scale behind urban service provision were general [15, 16], both deprivations should fall with city size. The everyday regime behaves as predicted (Fig. 3A): the log-log elasticity of mean everyday deprivation on population is -0.196 (country-clustered SE 0.023, $p_{\mathrm{wild}} = 0.0002$), a fall of roughly 19 % per e-fold of population, and size alone explains 44 % of the cross-city variance. The median walk to the nearest facility of each everyday category is 4–8 minutes, and larger cities carry denser networks. The estimate is stable under country fixed effects, leave-one-out deletion, robust estimators, exclusion of the desert capitals and a mapped-facility-density control (SI Appendix).

The emergency regime breaks the pattern. The elasticity of mean emergency deprivation cost is -0.059 ($p_{\mathrm{wild}} = 0.21$) and size explains 2 % of the variance. Non-significance alone would be weak evidence, so the null is bounded: two one-sided equivalence tests place the elasticity within ± 0.13 [35], and the minimum detectable elasticity at 80 % power is 0.12, so the design could have found a gradient of the everyday magnitude. The claim is "no detectable scaling, bounded within ± 0.13", never "proven flat". The median FUA has 6 emergency departments and 6 ambulance stations for its whole territory, a median drive of 12 minutes to either, and the facility count of a national emergency system does not multiply with population the way pharmacies do. A paired regime test within cities puts the everyday slope 0.137 below the emergency slope ($p_{\mathrm{wild}} = 0.002$): agglomeration benefits are regime-specific, and no single exponent describes what size does to service deprivation. Size is not irrelevant to emergency deprivation, however: a quadratic term is significant (wild $p = 0.001$), an inverted U in which the middle strata carry the highest cost and the four largest FUAs the lowest; with four cities in the top stratum we report the shape and do not headline it (SI Appendix).

Within-city inequality moves the other way (Fig. 3B). The Gini of everyday deprivation rises with population (elasticity $+0.062$, $p_{\mathrm{wild}} = 0.0012$); the emergency Gini is size-flat at the baseline ($+0.004$, $p_{\mathrm{wild}} = 0.84$); the paired difference of -0.058 is significant ($p_{\mathrm{wild}} = 0.029$). Size

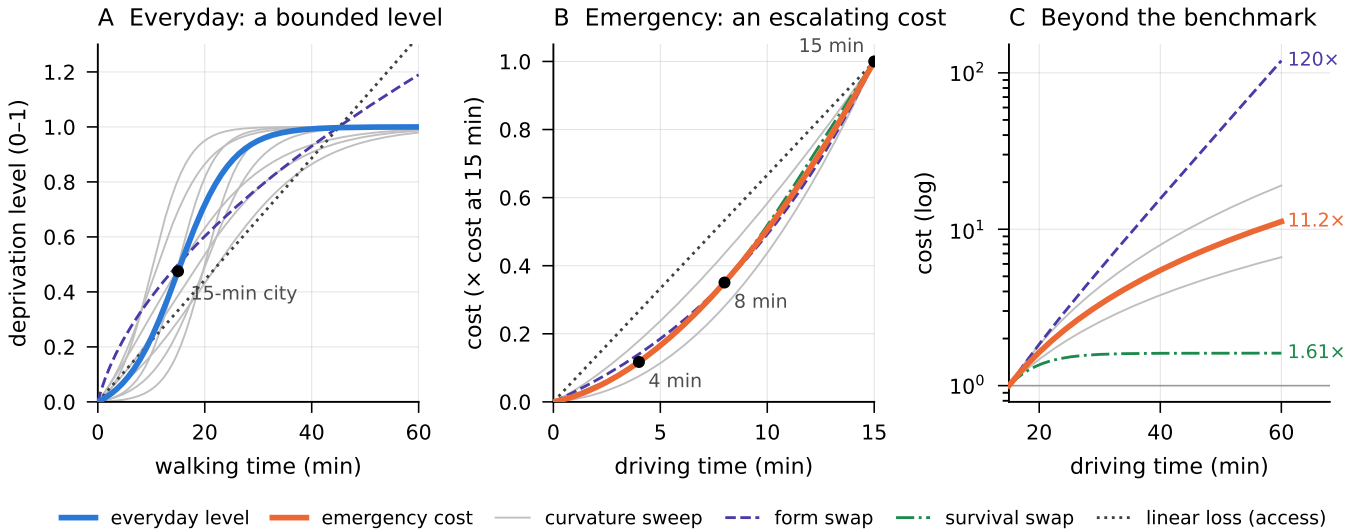


Figure 1: The two deprivation functions. (A) Everyday deprivation level against walking time: bounded, saturating, inflection at the 15-minute anchor. (B) Emergency deprivation cost against driving time inside the benchmark window, in multiples of the cost at the 15-minute EMS target; all forms coincide at the clinical anchors (4, 8, 15 min) by construction. (C) Beyond the target no benchmark exists and three escalation hypotheses diverge (log scale). Grey: curvature sweep; dashed: form swaps (SI Appendix); dotted: the linear loss a travel-time average implies. Panels are on different scales.

buys lower mean everyday deprivation at the price of a wider spread around it, while emergency inequality is set elsewhere. The everyday slope is positive under all 14 deprivation parameterisations and significant in 13, the exception being the concave form swap (+0.026, $p = 0.083$) [36]. The emergency half is conditional, and the conditionality is a finding: beyond the 15-minute benchmark no clinical anchor exists, and there the three literature-grounded escalation forms diverge (Fig. 1C). Under the bounded survival curve and under the polynomial baseline the emergency Gini is size-flat, in agreement with the deprivation-free travel-time Gini (−0.015, $p = 0.42$); under exponential escalation it rises (+0.098, $p = 0.0003$), because exponential pricing turns the Gini into an index of extreme-tail depth (SI Appendix). Two statements hold under every parameterisation: everyday inequality rises with size, and emergency inequality never falls with it.

Emergency deprivation follows national emergency-system coverage, not city size

City size explains 2 % of the variance in mean emergency deprivation cost; 78 % lies at the country level (25 % between the four macro-regions, 54 % between countries within regions), and the country level also carries 70–76 % of the variance in both Ginis and their gap (SI Appendix). We first identify the structure in the city vectors, then its geography, then test it at the level at which it varies.

The 67 cities do not fall into types. k-means clustering on the full standardised city vector separates 62 cities from 5 (silhouette 0.65, bootstrap ARI 0.87, Gaussian-null $p = 0.001$), an outlier group with no agreement with the region partition (ARI 0.001): the five desert capitals, separated by their beyond-threshold emergency travel-time tails. Ward clustering cuts the same axis more conservatively (65/2), and re-clustering without the outliers finds one further cut along it, 48/14, isolating partial-desert cities with real emergency tails (Oslo, Stockholm, Warszawa, Sofia, Zagreb, Palermo and Luxembourg among them). The nesting is evidence for a gradient, not types: a one-dimensional *coverage ordering* of the emergency periphery, covered (48), partial desert (14), desert (5), robust to the feature set (ARI 1.00 and 0.97) with a soft boundary between covered and partial (SI Appendix). A city's grade expresses how much of its population lives beyond the emergency response benchmark. The grade, and no other variable we test, sets the emergency level: in a country-clustered model with centred log population, the grade dummies shift mean emergency deprivation cost overwhelmingly (Wald $p \approx 10^{-15}$; median 1.02, 1.87 and 3.37 multiples of g(15) across the three grades), while the slope-by-grade interaction is boundary-sensitive and not headlined.

The ordering has a geography (Fig. 4A). All five deserts are capitals of Central-Eastern European (CEE) countries; the partial-desert grade is a Nordic and CEE phenomenon (7 of 13 Nordic and 5 of 20 CEE cities, against 1 of 16 in the West and 1 of 18 in the South); West and South are almost entirely covered. Grade and region are different partitions and related: the emergency tail is a feature of the sparse peripheries of Nordic FUAs and of the national emergency systems of Central-Eastern Europe. In the inequality plane (Fig. 4B) the deserts are the five highest points on the emergency-Gini axis and the partial deserts fill the band below them, so a city's

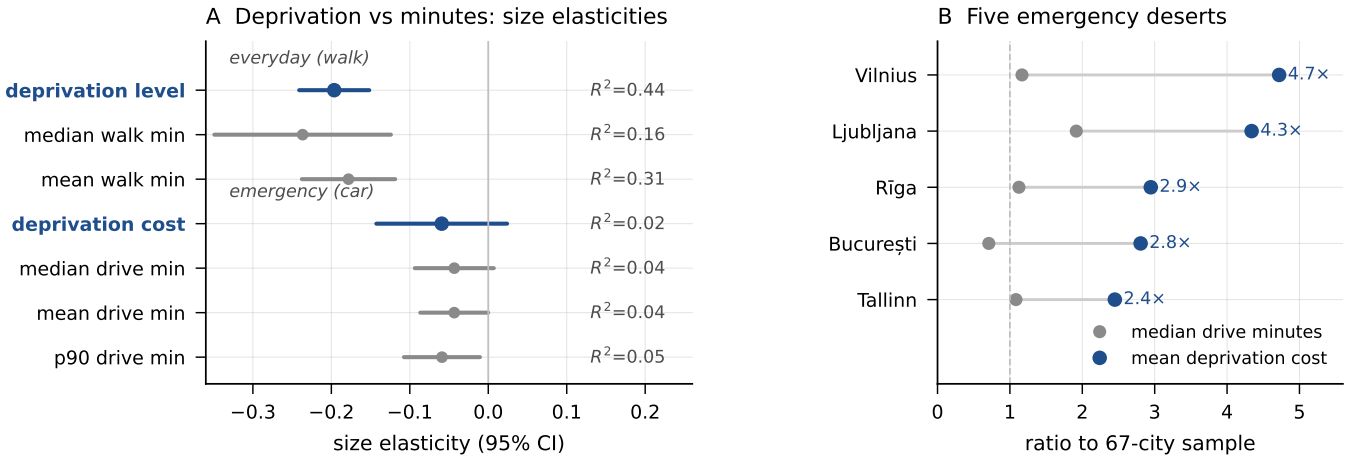


Figure 2: Deprivation against access. (A) Size elasticities of deprivation-based (blue) and minutes-based (grey) outcomes per regime (95 % CIs, country-clustered). (B) The five emergency-desert capitals: ratio to the 67-city sample under the access average (median drive minutes) and under mean emergency deprivation cost. Sources: SI Appendix Tables.

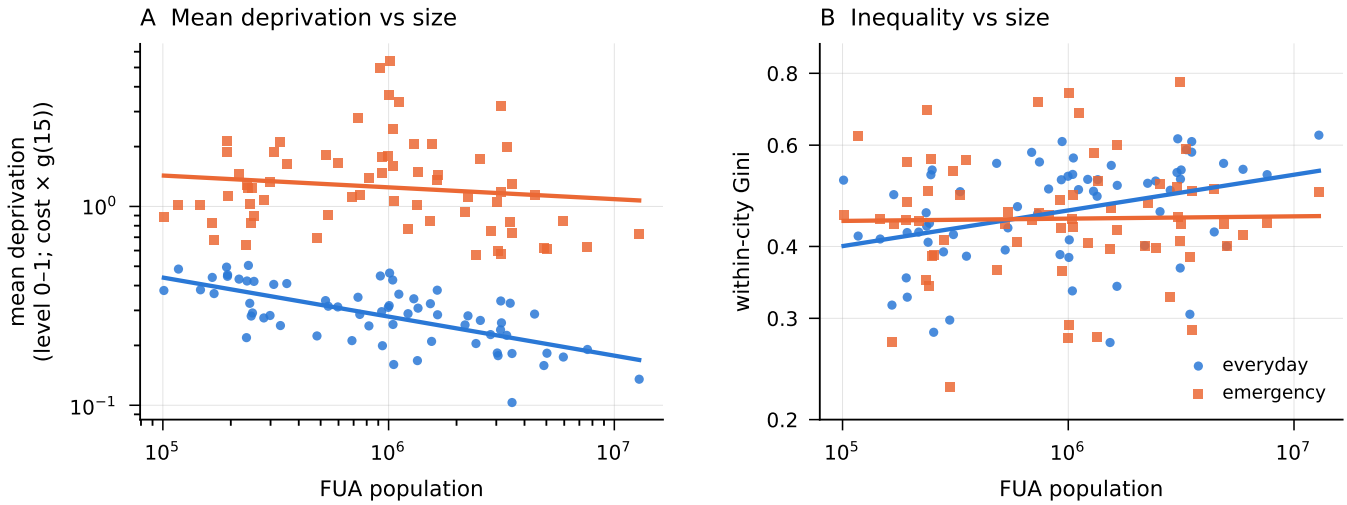


Figure 3: What city size does to deprivation, per regime (log-log; 67 FUAs; cross-sectional). (A) Mean deprivation: everyday elasticity −0.196 (wild $p = 0.0002$, $R^2 = 0.44$); emergency −0.059 ($p = 0.21$, bounded within ±0.13); paired difference +0.137 ($p = 0.002$). The two series are in different units and only the slopes compare. (B) Within-city Gini: everyday elasticity +0.062 ($p = 0.0012$); emergency +0.004 ($p = 0.84$); paired difference −0.058 ($p = 0.029$).

emergency-side inequality is, to a first approximation, its coverage grade. Across cities the two Ginis are uncorrelated ($r = 0.12$, $p = 0.32$): how unequally a city distributes everyday deprivation says nothing about how unequally it distributes emergency deprivation, the cross-city face of the two-regime claim.

Because macro-region is a property of countries, whole countries are permuted across regions in every regional test. The emergency Gini differs by region (Cohen $f = 0.74$, $p_{\text{perm}} = 0.0024$), as do the divergence gap ($p_{\text{perm}} = 0.010$), the coupling ρ of the two surfaces ($p_{\text{perm}} = 0.0015$) and both compounding measures ($p_{\text{perm}} = 0.0008$, $p_{\text{perm}} = 0.0023$). Central-Eastern Europe compounds: the highest emergency inequality (median Gini 0.54) with the strongest coupling (median $\rho = 0.55$). Every CEE contrast survives re-testing without the five deserts, and CEE's dispersion of the emergency Gini is about twice any other region's ($p_{\text{perm}} = 0.028$). The everyday Gini's apparent regional contrast does not survive country-level testing ($p_{\text{perm}} = 0.27$): the Southern cities split by country, the Spanish cities and Athina at 0.50–0.61 against the Italian and Portuguese ones at 0.27–0.42, which coincides with a known mapping artefact in general-practitioner tagging (Limitations). The paper therefore makes no regional everyday-Gini claim and no cross-country comparison of everyday levels.

Compounding is the norm, and children carry it

The co-location typology cuts both percentile surfaces at their population-weighted medians, which fixes the

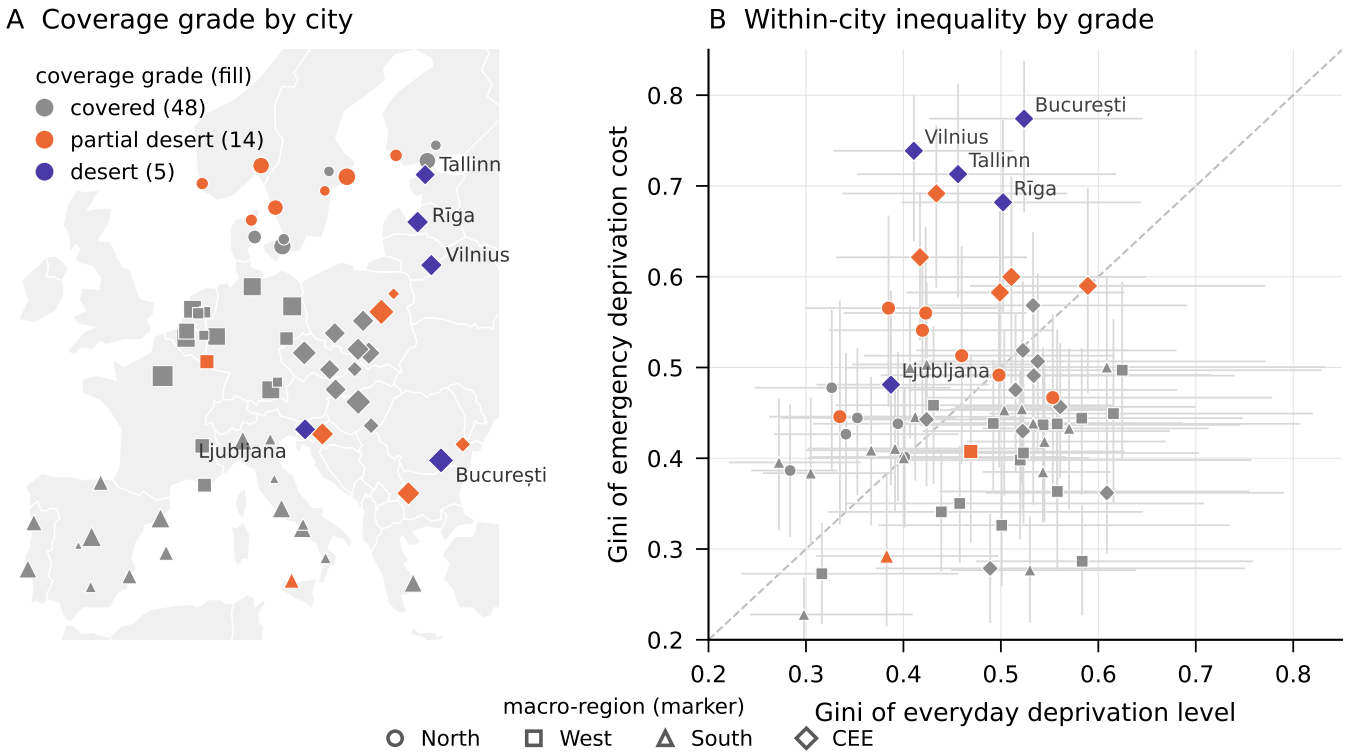


Figure 4: The geography of emergency coverage. (A) The 67 FUAs by coverage grade (fill) and macro-region (marker), sized by population; the five deserts are labelled. (B) Within-city Gini of the emergency deprivation cost against the everyday level, same fill and markers; whiskers span the min–max envelope of each Gini over the deprivation-curvature sweep.

marginals at one half; the 2×2 class table then has a single free number, the high-high (HH) compounding share, and statistical independence corresponds to HH = 25 %. The observed share averages 33 % and exceeds independence in 66 of 67 cities; the rank coupling ρ spans −0.02 to 0.74. Compounding is not an exotic overlap of two rare conditions; across urban Europe it is the default relationship between the two deprivations. The "how high is high" threshold moves the HH share 21 times more than the whole deprivation calibration does, so the headline compounding measure is the threshold-free intensity, the population-weighted mean depth to which the typical resident sits inside both percentile distributions (1/3 under independence); it shows the same regional structure. Spatially, compounding concentrates in peripheries where the saturated everyday level meets the escalating emergency cost, while dense cores hold most population in the low-low class (all 67 maps in SI Appendix; per-cell classes carry a 24 % routing-engine flip share and are read as patterns only).

Who lives there? Following the social-vulnerability tradition [27], EU-harmonised census age strata are pooled across all 67 cities, each compared against the cells where its own census variable is published; national fine-grid strata are per-city depth and never pooled. The result is one-sided (Fig. 5): cells in the top quartile of child share carry more everyday deprivation than their comparison cells in 88 % of cities (median ratio 1.27) and more compounding in 88 % (median HH-share gap +0.13), strongest in Central-Eastern Europe (1.44) and the North (1.30). The elderly show no such pattern at the European level (median ratio 0.96, above parity in 37 % of cities), a null that is itself a result; national data resolve sharper per-city patterns (in Hamburg, elderly 1.37, children 1.52, low-rent 1.61) but stay per-city. The European claim is about children: whatever sorting mechanism places families with children in the compounded periphery operates across nearly the whole sample, in a group whose daily radius is short and whose emergency risk is real.

Discussion

Deprivation as the welfare measure of urban proximity

The first contribution is to bring deprivation theory from humanitarian logistics [17, 18] into the analysis of a functioning city. Where results are rank-based, travel time and deprivation agree almost perfectly, which certifies the accessibility literature's rank-level findings [13, 14] at continental scale. Where results concern levels, inequality magnitudes or tails, travel time was never a welfare measure, and the divergence is not noise: the desert

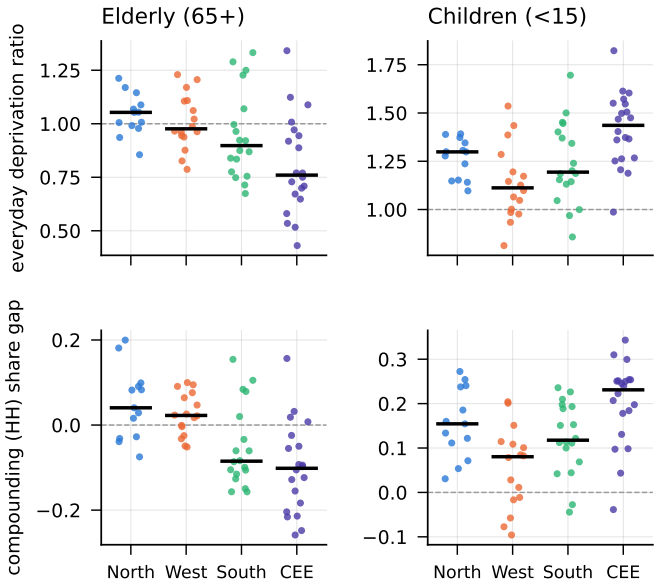


Figure 5: Who carries the deprivation. Census-harmonised strata by macro-region, pooled across 67 cities against covered-cell references (dashed: parity; bar: regional median). Top: everyday deprivation ratio; bottom: compounding (HH) share gap. Children: higher everyday deprivation in 88 % of cities (median 1.27) and higher compounding in 88 % (median gap +0.13). Elderly: parity (0.96); no European elderly claim is made.

capitals sit at 0.71–1.92 times the sample by access and 2.45–4.72 times it by deprivation, and the choice of measure decides whether an emergency-side size story exists at all. The same limitation is conceded from inside the resilience literature, where the latest building-level frameworks measure access as network distance to the nearest facility and note that distance captures one dimension of access only [37]. The theory also explains why the two regimes are different objects: a bounded level for substitutable needs and an unbounded cost for time-critical needs are what it implies for the two kinds of good, and the clinical response literature [8, 23] supplies the anchors that a value of time estimated on reschedulable trips cannot. The framework travels to any domain with anchors of comparable strength.

What urban growth delivers, regime by regime

Scaling theory rests on exponents that hold across cities and quantities [15, 16]; its refinements condition them on the stage of a national urban system [32]. Our results add a different condition: the exponent depends on the regime of the service class, within one urban system and one measurement. Everyday deprivation scales (-0.196, 44 % of variance); emergency deprivation does not (bounded within ± 0.13); the two slopes differ within cities. A single infrastructure exponent is not a sufficient statistic for what size does to service deprivation, and in a mature urban system, where large cities no longer grow faster than small ones [32], what size still delivers is lower everyday deprivation. The inequality result cuts deeper: everyday inequality rises with size under every parameterisation, so the agglomeration that lowers mean everyday deprivation widens its spread, which argues that the scaling of within-city, welfare-weighted distributions is where the next generation of scaling laws has to live. The 15-minute-city paradigm [2, 3] emerges recast: its threshold anchors the everyday regime well, and precisely because that regime saturates, a proximity paradigm is an everyday-services paradigm, silent about the escalating-cost regime and about the populations where both fail.

Built for the everyday, tested by the emergency

Resilience research defines resilience through equitable access to essential services [24, 25], finds that access equality and network resilience reinforce each other [38], and shows everyday vulnerability failing to predict crisis capability [26]. Our two-regime measurement gives that distinction quantitative form across an urban system: the everyday surface is set by the city and its density, the emergency surface by the coverage of a national emergency system, so a city's resilience cannot be read from either alone, and a city built for the everyday regime is not thereby resilient. Compounding is the case the coupled-systems argument predicts: the same neighbourhoods in the periphery of both systems, above independence in 66 of 67 cities, strongest where emergency inequality is highest, and consistently the neighbourhoods with many children. The measurement is of the emergency regime under normal conditions, the floor from which any disruption starts. Frameworks that predict building-level loss of access after a hazard take a pre-hazard baseline of network distance to the nearest operational facility as their starting point [37]; the emergency-deprivation geography measured here is that baseline for urban Europe, in welfare terms, and it matters more each year as heat exposure and heat mortality rise [4, 5] and the Preparedness Union Strategy asks member states to plan hospitals, schools and transport for the emergency regime [9]. The practical reading follows the two jurisdictions. Everyday inequality is a big-city problem: it rises with size, and intra-urban service planning is the lever. Emergency deprivation is a national coverage problem: it follows coverage grade rather than size, and its extreme cases are five capitals whose peripheries no municipal instrument can serve alone; the coverage ordering is a screening variable access averages do not provide. The consistent carriers of compounded deprivation are families with children, which makes school- and family-anchored service planning a compounding remedy rather than a sectoral one.

Limitations

Every size gradient is cross-sectional. The analysis covers walking for everyday services and driving for emergency care; public transit is out of scope by design. Free-flow civilian drive times anchor the emergency cost scale and are not response-time predictions; congestion lengthens drive times most in the largest cities, so free-flow routing understates emergency deprivation there and pushes the emergency elasticity toward a size benefit, which makes the absence of a gradient a conservative finding. Disruption itself is not modelled, so the emergency surface is the pre-disruption baseline that hazard-specific frameworks perturb [37]. Census strata are 1 km attributes broadcast to the 100 m grid. Per-cell typology classes carry a roughly 24 % routing-engine flip share and are read only as patterns. The deprivation layer itself is the first thing a reader should attack, and SI Appendix reports the attack in full: levels move with the calibration and are meant to, while ranks and classes do not. Mapped general-practitioner density varies about 12-fold across countries in a way that reflects OpenStreetMap tagging conventions rather than health systems, which overstates everyday deprivation levels in the Nordic and Southern countries; cross-country everyday level comparisons are therefore not supported, while the size gradients are unaffected (mapped density is uncorrelated with size, and a density control leaves the elasticities in place). Eight cities have only one of the two emergency services mapped; none is in the desert group and no headline turns on it (SI Appendix).

Materials and Methods

Design and sample

The study is a cross-sectional comparison of 67 FUAs in 24 countries under one harmonised city definition, the Eurostat–OECD functional urban area [28], with trajectories read from the cross-sectional size gradient as in [32]; there is no longitudinal component. The sample is a stratified draw over four macro-regions and four population strata (100,000–250,000; 250,000–1 million; 1–5 million; above 5 million; 15/18/30/4 cities), seeded once and fixed before any surface was computed. Countries nest inside regions, and every cross-city test is built around that nesting.

Data and travel times

The analysis unit is the populated 100 m cell of the GHS-POP R2023A grid [39], clipped to each FUA; every aggregate is population-weighted. Facilities come from OpenStreetMap under one tag taxonomy: five everyday categories reached on foot (general practitioners, pharmacies, supermarkets, schools, green spaces) and two emergency services reached by car (emergency departments, ambulance stations); median counts per FUA are 93, 228, 325, 409, 636, 6 and 6. Travel times are routed with R5 [40] over the OpenStreetMap network, walking at 4.8 km/h with a 30-minute cut-off and driving at free-flow speed with a 60-minute cut-off, from facilities to cells and transposed (median car asymmetry one minute; the transposed direction is the correct one for ambulance response). Cells with no network path are masked under one shared mask; cells that are routable but beyond a service's cut-off receive a large finite time so that the everyday level saturates and the cell stays on the map. Census strata come from the Eurostat Census 2021 1 km grid (harmonised age groups) and, where published, national fine grids; withheld counts are missing, never zero. A friction-surface engine [12] is a declared sensitivity variant after a two-city cross-check showed city-specific level bias (SI Appendix).

Deprivation functions

In both regimes the functional form is transferred from the estimated literature and the curvature calibrated to domain anchors, because the published coefficients were estimated on hours of deprivation and do not transfer to intra-urban minutes. For each everyday service, travel times to reachable facilities are adjusted for congestion by a two-step floating catchment area factor [41, 42] and reduced by a soft minimum, $t_{is}^{\mathrm{eff}} = -\kappa^{-1} \ln \sum_j \exp(-\kappa\, t_{ij} c_j)$, so that several similar options count as slightly closer than the nearest alone. The everyday deprivation level is the logistic

$$D_{is} = \left[1+\exp(-k\,(t_{is}^{\mathrm{eff}} - t_0))\right]^{-1}, \quad t_0 = 15\,\mathrm{min},\ k = 0.2\,\mathrm{min}^{-1}, \tag{1}$$

the saturating S-curve of needs-based severity [19] with its inflection at the 15-minute-city threshold [2], averaged over the service categories reachable from the cell. The emergency deprivation cost uses the nearest facility and a convex Box-Cox function [20, 22],

$$C_i = \frac{g(\min_j t_{ij})}{g(15)}, \qquad g(t) = \frac{(t+1)^{\lambda} - 1}{\lambda}, \quad \lambda = 1.8, \tag{2}$$

anchored on the EMS response benchmarks: negligible below the 3–4 minute ideal [29, 43], about a third of the benchmark cost at the contested 8-minute standard [29, 30, 44], and equal to it at the 15-minute upper target [31], with convexity supported by distance-mortality evidence [23]. Solving λ from the anchors gives 1.89–1.95, inside the sensitivity sweep. Division by $g(15)$ reports costs in multiples of the benchmark deprivation and leaves every rank-based statistic and both Ginis unchanged.

Indicators and cross-regime comparison

Only unit-invariant statistics cross regimes: elasticities, the population-weighted within-city Gini (mean-normalised, so comparing the inequality of a level with that of a cost is the move of comparing an income with a wealth Gini) and rank statistics on the population-weighted mid-rank percentile of each surface. The co-location typology cuts both percentile surfaces at their medians; at that split HL = LH = 0.5 − HH and LL = HH, so the HH share is the one free number and independence gives 0.25. Alongside it we report the Spearman coupling ρ of the two percentile surfaces and the threshold-free compounding intensity, the population-weighted mean of $\min(p_i^{\mathrm{ev}}, p_i^{\mathrm{em}})$, which equals 1/3 under independence and 1/2 under perfect coupling. Vulnerability strata (top quartile of child share, of share aged 65 and over) are compared against the cells where their own variable is published; strata covering less than half a city's population are dropped from the pooled summary.

Inference

Each outcome is regressed as $\ln y_c = \alpha + \beta \ln P_c + \varepsilon_c$ [15], with standard errors clustered on country and a wild cluster bootstrap p-value with the null imposed (Rademacher weights, 4,999 draws) [45, 46]. The emergency null is bounded by two one-sided equivalence tests [35]; the regime difference is tested within cities by a regime × log-population interaction; every headline elasticity carries leave-one-out, Cook's-distance, Huber and median-regression diagnostics, a mapped-facility-density control and quadratic and stratum-dummy non-linearity checks. Regional contrasts compare country means and permute whole countries across regions (10,000 permutations); the city-level ANOVA is an anti-conservative reference. A nested sum-of-squares decomposition, a Fligner–Killeen dispersion test under the same permutation, and re-tests without the desert capitals characterise where the variance lies. Clustering runs k-means and Ward on the robust-scaled city vector with k selected by silhouette, each partition carrying silhouette, bootstrap adjusted Rand index, a Gaussian silhouette null and the ARI against the region partition; when the winning split isolates an outlier group a peeled second pass re-clusters the rest, and the coverage grade is read off the two passes, never from a hand-set threshold. A structured sensitivity sweep varies curvature ($k \in \{0.1, 0.2, 0.3\}$, $t_0 \in \{10, 15, 20\}$, $\lambda \in \{1.4, 1.8, 2.2\}$), swaps the functional forms while pinning the same anchors, sweeps the accessibility model and the typology threshold, and reports rank agreement, envelopes, flip-cell shares and the specification curve of the inequality slopes over all 14 parameterisations [36]. Full procedures, including the reachability split, the robustness layers and the routing-engine cross-check, are in SI Appendix.

Data and code availability

All tables and figures trace to the persisted 67-city state; every quoted number is checked against its source table by an automated audit of 190 checks run in continuous integration. Code, configuration, result tables, per-city maps and sensitivity sweeps are available in the study repository and its results branch. Population grid: GHS-POP R2023A. Facilities and networks: OpenStreetMap. Census strata: Eurostat Census 2021 grid; German Zensus 2022 grid. Boundaries: Eurostat–OECD functional urban areas.

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